Assignment 3

Course: *Big Data*
Due date: *April 14th*

This is the first part of the final project, so you should be working with two students in groups. Use Python and Dask with any additional package of your choice.

Data

We will explore the New York City Open Data - TLC Trip Record Data for Yellow Taxis. The data is available at `https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page`.
Or on the ARNES cluster: `/d/hpc/projects/FRI/bigdata/data/Taxi`.
For this homework, you can work on a subset of the data.

Data ingestion.

Import the parquet files and store data:

- for 5 years in *CSV* format
- for 1 year in both *HDF*5 and *CSV* formats.
- in new parquet files with more optimal partitioning. Use the $dataframe.to\_parquet()$ function. Partition data by year (see $partition\_on$) and use $row - group$ to create evenly sized chunks of data.

Be careful to match data from different years (columns for different data should be represented the same). Take care of both column names and data types, accounting for missing values. Try to parallelize your work if possible.

Compare different sizes of the data in different formats.

Comment on the results.

Augment the original data with sources of additional information:

- Weather information (based on pickup/dropoff date-time).
- Vicinity of different schools and universities (based on pickup/dropoff location).
- Major events in the vicinity (based on pickup/dropoff date-time and location).
- Vicinity of major businesses and attractions (based on pickup/dropoff date-time).

You can find many (but not all) sources in the New York City Open Data Repository (`https://opendata.cityofnewyork.us/`). You can also find a lot of New York data online, including weather data.

Connect to the Arnes cluster and run some code.

Refer to the instructions of assignment "Arnes cluster".